Sales and inventory analysis

DAT 300 | Academic coursework | Power BI analysis and gap assessment

An academic analysis of course-provided sales and inventory data, followed by a proposal for better-connected reporting. The strongest result is a clear boundary between questions the available data can answer and inventory decisions that need more evidence.

Business questions

Which periods have stronger sales, which products generate the most revenue and profit, and is the inventory data sufficient for planning?

Work documented in the reports

- Created Power BI visuals for sales over time, revenue by product and inventory balances. The later gap analysis also included profit margin and total profit comparisons.
- Reviewed field relevance, date and product consistency, missing information, and the limits of combining the two sources.

Selected findings

- Staples led the preliminary revenue comparison. The 3-ring staple pack led the later total-profit comparison. These are different measures.
- The gap-analysis report identified November, September and December as the strongest sales months in its month-level comparison. This is a descriptive result, not a demand forecast.
- Inventory covered only Bike and Car from January through May 2020. Both balances were lower by May; this small extract cannot support inventory conclusions for the full sales catalog.

Recommendation

Collect inventory coverage for key products, reorder levels, stockouts, replenishment dates and product costs. Standardize product and date fields, validate the model and test reporting before operational use.

Scope and evidence

The reports contain Power BI screenshots. No PBIX file, underlying model, DAX definitions or executable workflow was available for this review. ETL, broader integration and production reporting were proposed. No business impact or deployed dashboard is claimed.

Skills demonstrated

Power BI visualization; data validation; business analysis; gap analysis.

Source and attribution

Olga Solodovnikova, DAT 300 Project One Preliminary Data Assessment and Project Two Gap Analysis Proposal, Southern New Hampshire University. Source data credited in both reports to uCertify sales and inventory course datasets.